



Comparative genomics with a multidrug-resistant *Klebsiella pneumoniae* isolate reveals the panorama of unexplored diversity in Northeast Brazil

Rodrigo Profeta^a, Núbia Seyffert^{a,b}, Sandeep Tiwari^a, Marcus V.C. Viana^a,
Arun Kumar Jaiswal^{a,c}, Ana Carolina Caetano^a, Daniel Henrique Bucker^a,
Luciana Tavares de Oliveira^d, Roselane Santos^a, Alfonso Gala-Garcia^a, Rodrigo B. Kato^a,
Francine F. Padilha^e, Isabel B. Lima-Verde^f, Preetam Ghosh^g, Debmalaya Barh^h,
Aristóteles Góes-Netoⁱ, Henrique C.P. Figueiredo^j, Siomar C. Soares^{a,c}, Roberto Meyer^b,
Bertram Brenig^k, Pablo I.P. Ramos^l, Vasco Azevedo^{a,1}, Thiago L.P. Castro^{a,b,*,1}

^a Department of Genetics, Ecology and Evolution (GEE), Federal University of Minas Gerais, Belo Horizonte, Minas Gerais, Brazil

^b Institute of Health Sciences, Federal University of Bahia, Salvador, BA, Brazil

^c Department of Microbiology, Immunology and Parasitology, Institute of Biological and Natural Sciences, Federal University of Triângulo Mineiro, Uberaba, MG, Brazil

^d Laboratory of Clinical Pathology, Hospital das Clínicas, Federal University of Minas Gerais, Belo Horizonte, Minas Gerais, Brazil

^e Tiradentes University, Technology and Research Institute, Biomaterials Laboratory, Aracaju, Sergipe, Brazil

^f Swedish University of Agricultural Sciences, Uppsala, Sweden

^g Department of Computer Science, Virginia Commonwealth University, Richmond, VA 23284, USA

^h Institute of Integrative Omics and Applied Biotechnology (IIIOAB), Nonakuri, Purba Medinipur, West Bengal 721137, India

ⁱ Molecular and Computational Biology of Fungi Laboratory, Department of Microbiology, Institute of Biological Sciences, Federal University of Minas Gerais, Belo Horizonte, Minas Gerais, Brazil

^j AQUACEN – National Reference Laboratory of Aquatic Animal Disease, Ministry of Agriculture, Livestock and Food Supply, Veterinary School, Federal University of Minas Gerais, Belo Horizonte, MG, Brazil

^k Institute of Veterinary Medicine, University of Göttingen, Burckhardtweg 2, Göttingen, Germany

^l Gonçalo Moniz Institute, Oswaldo Cruz Foundation, Salvador, Bahia, Brazil

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ABSTRACT

The emergence of community acquired infections increases the public health concern on *K. pneumoniae* and closely related bacteria among which antimicrobial resistance spreads. We report a multidrug-resistant *K. pneumoniae* isolate, B31, of a patient infected in the community and admitted to an intensive care unit in Northeast Brazil. Antimicrobial susceptibility and genome information were thoroughly investigated to characterize B31 in front of 172 sequenced strains of different countries. Assigned to the Sequence Type 15, which is globally spread, B31 presented extended spectrum beta-lactamase, tigecycline and ciprofloxacin resistance. Genome sequencing revealed most resistance genes being carried by plasmids with high dissemination potential. The absence of main virulence factors, like yersiniabactin and colibactin, apparently suggests a mild pathogenic strain which, on the contrary, persisted and caused severe infection in a previously healthy patient. The present study contributes to unveil the unclear genomic scenario of virulent and multidrug-resistant *K. pneumoniae* in Brazil.

Abbreviations: MDR, multi-drug resistant; AST, antimicrobial susceptibility test; NCBI, National Center for Biotechnology Information; MLST, Multilocus Sequence Typing; ST, Sequence Type; KL, capsule types; WAG, Whelan And Goldman; iTOL, Interactive Tree Of Life; ICU, Intensive Care Unit; Sul, Sulfonamide; Tet, Tetracycline; Tmt, Trimethoprim; RI, resistance islands; PI, pathogenicity islands; MIS, miscellaneous islands; T6SS, Type VI Secretion System; ANI, Average Nucleotide Identity.

* Corresponding author at: Department of Genetics, Ecology and Evolution (GEE), Federal University of Minas Gerais, Belo Horizonte, Minas Gerais, Brazil.

E-mail address: thiago.lui@ufba.br (T.L.P. Castro).

¹ These authors share senior authorship.

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1. Introduction

Klebsiella pneumoniae is a Gram-negative, non-motile, oxidase-negative and lactose fermenting Enterobacteriaceae (Ramos et al., 2014). Often surrounded by a capsule, which confers a viscid consistency to bacterial colonies, the cells are rod-shaped and may be arranged in pairs or short chains (Drancourt, 2015). *K. pneumoniae* persists in various ecological niches, ranging from soil to water (Podschun and Ullmann, 1998; Podschun et al., 2001), and is one of the most important opportunistic pathogens of medical interest (Struve and Krogfelt, 2004). In healthcare settings, it can be acquired via person-to-person contact or, less frequently, through contamination of the environment (Siegel et al., 2007).

Although it is naturally present in 3.8 % and 9.5 % of healthy stool and nasal samples, respectively (Conlan et al., 2012;7.); *K. pneumoniae* is commonly associated with nosocomial infections (Magill et al., 2018;379:1732–44.). *K. pneumoniae* may cause bloodstream infections and affect the lungs, skin pharynx, and gastrointestinal and urinary tracts. The high incidence of antibiotic-resistant strains makes *K. pneumoniae* one of the most common causes of hospital-acquired infections worldwide (Martin and Bachman, 2018; Dorman and Short, 2017). The correct identification of *K. pneumoniae*; phylogroup KpI, is difficulted by its close relationship with the phylogroups KpII (*K. quasipneumoniae*) and KpIII (*K. variicola*) (Holt et al., 2015). Conventional biochemical tests are inefficient to distinguish isolates of these phylogroups (Alves et al., 2006).

Virulence determinants such as the polysaccharide capsule and siderophores support the establishment of infection and are present in many *Klebsiella* species (Wyres et al., 2016; Lam et al.:098178., 2018). Appropriate bacterial identification, typing and screening of resistance and virulence determinants are fundamental for the adoption of more targeted interventions to control emerging microbial threats such as the multi-drug resistant (MDR) *K. pneumoniae* (Wyres et al., 2016). Whole-genome sequencing has been a great ally to provide accurate diagnosis in patients with symptoms commonly caused by a variety of bacteria and unveil the genetic basis of drug resistance and virulence in human pathogens (Seki et al., 2013).

In the present study, we report the isolation of an MDR *K. pneumoniae* strain from a patient admitted to a health care unit in Northeast Brazil. Initially, the suspicion was that the infection had been caused by other species commonly associated with pneumoniae (e.g. *Staphylococcus* or *Streptococcus* spp.) or *Corynebacterium pseudotuberculosis*, which may affect the lungs and had been recently manipulated by the patient in a research institute. Although several reports of human *K. pneumoniae* infections have been made in the last years, we provide comprehensive phenotypical and genotypical characterization for this strain as hardly seen in Brazil. In addition, this is the first work to extensively characterize a *K. pneumoniae* isolate from Northeast Brazil, providing comparative analyses over isolates from different parts of the world and information that might be useful for population genomics of this bacterium in the country.

2. Methods

2.1. *K. pneumoniae* isolation and characterization

K. pneumoniae B31 was isolated in 2016 from the bronchoalveolar lavage of an infected patient with airway symptoms in the state of Sergipe, Northeast Brazil. A bronchoalveolar lavage sample obtained from the patient was submitted to serial dilutions in steam sterilized 0.9 % sodium chloride solution and plated on Columbia Blood Agar (Oxoid) and Brain Heart Infusion (HiMedia) agar plates, which were incubated for 24 h at 37 °C. Organism identification was conducted using MALDI Biotyper v. 3.0 (Bruker Daltonics, Germany) (Santos et al., 2013) and

antimicrobial susceptibility tests (AST) were performed using Vitek (bioMérieux), according to the manufacturers' recommendations.

2.2. Genome sequencing

B31 DNA was obtained through mechanical lysis of cells harvested from culture broth, as described (de Aguiar et al., 2016). A 450 bp read library was constructed using NEBNext Fast DNA Fragmentation and Library Preparation Kit (New England Biolabs, Ipswich, NE, USA) and evaluated using Agilent 2100 Bioanalyzer. Whole genome sequence was obtained using Illumina HiSeq 2500 (Illumina, San Diego, CA, USA). Quality of DNA reads was assessed using FastQC (FastQC, 2010). *De novo* assembly was performed using SPAdes (v3.9.1) (Bankevich et al., 2012). The best k-mer length was estimated using KmerGenie (v1.7044) (Chikhi and Medvedev, 2014). Statistics about the quality of the assembly were generated using QUAST (Gurevich et al., 2013). *K. pneumoniae* Kp_Goe-39795 genome sequence from NCBI (National Center for Biotechnology Information) (accession CP018458.1) was used as the reference to generate the B31 genome scaffold in CONTIGuator. Plasmid sequences were screened using PlasmidFinder (v1.3) (Carattoli et al., 2014) and plasmid assemblies obtained using PlasmidSpades (Antipov et al., 2016). Plasmid contigs were checked using BLASTn against the non-redundant database of NCBI. Genome and plasmid sequence gaps were filled using FGAP (Piro et al., 2014) and CLC Genomics Workbench (v6.5) (<https://www.qiagenbioinformatics.com/>). Annotation was performed using the RASTtk (Rapid Annotation Using Subsystem Technology) (Brettin et al., 2015) annotation service available in PATRIC (Pathosystems Resource Integration Center) (Wattam et al., 2017). Annotation of Kp_Goe-39795 genome was transferred to B31 using an in-house script. Comparison between annotations allows identifying frameshifts in coding sequences. Manual curation of frameshifts was conducted using CLC Genomics Workbench and Artemis v16.0.049. Conserved mutations in frameshifted sequences were checked using BLASTn against the non-redundant database of NCBI. Nucleotide insertions/deletions caused by sequencing artifacts were corrected using BLASTp against the UniProt database (Apweiler et al., 2004).

2.3. Genome-based *K. pneumoniae* species validation

In silico confirmations of *K. pneumoniae* species were performed using JSpeciesWS to estimate, in an all-versus-all fashion, the Average Nucleotide Identities (ANI) between common coding regions in genomes (Richter et al., 2016). The ANIb (BLAST +) option was selected for the analysis. The type-strain genomes of *K. quasipneumoniae* subsp. *quasipneumoniae* 01A030, *K. quasipneumoniae* subsp. *similipneumoniae* 07A044, *K. variicola* DSM 15968, and *K. pneumoniae* subsp. *pneumoniae* DSM 30,104 were retrieved from GenomesDB and used as the references (Richter et al., 2008). The ANI value of 96 % was set as the threshold to define species boundaries. This percentage has been found to correlate well with DNA-DNA hybridization studies (Goris et al., 2007) and has been recommended by an NCBI-centered taxonomic workshop group (Federhen et al., 2016).

2.4. Genomic Multilocus sequence typing

Genomic Multilocus Sequence Typing (MLST) was conducted using seven housekeeping genes: *gapA*, *infB*, *mdh*, *pgi*, *phoE*, *rpoB*, and *tonB*. These genes encode glyceraldehyde 3-phosphate dehydrogenase, translation initiation factor IF-2, malate dehydrogenase, phosphoglucose isomerase, phosphorin E, DNA-directed RNA polymerase subunit beta, and periplasmic energy transducer, respectively. Sequence Type (ST) numbers were determined using *K. pneumoniae* BIGSdb (Bacterial Isolate Genome Sequence Database) (<http://bigsdb.pasteur.fr/Klebsiella/Klebsiella.html>). Kleborate (v2.0) (Lam et al., 2017) and the

Kaptive's reference database (Wyres et al., 2016) were used to assign the best matching capsule types (K-types) of *K. pneumoniae* strains.

2.5. Identification of virulence and drug resistance genes

Kleborate (v2.0) was used to screen virulence and drug resistance genes. This tool allows for the identification of genes *ybt* (yersinia-bactin), *clb* (colibactin), *iuc* (aerobactin), *iro* (salmochelin), and *rmpA*/*rmpA2* (related to the hypermucoid phenotype). Resistance genes were also screened using the ARG-ANNOT database (SRST2 version) (Gupta et al., 2014; Inouye et al., 2014). Pathogenicity and resistance islands in B31 were identified using GIPSY (Genomic Island Prediction Software) (Soares et al., 2016). The genome of *K. oxytoca* AR380, a non-pathogenic strain, was used as the reference for GIPSY. Prophages were screened using PHASTER (Arndt et al., 2016).

2.6. Comparative genomics for *K. pneumoniae*

The complete chromosome sequences of the available 172 *K. pneumoniae* strains Supplementary Table 1 were retrieved from GenBank (NCBI). PEPR (Phylogenomic Estimation with Progressive Refinement) (<https://github.com/enordber/pepr.git>) was used to extract the core proteome encoded by these genomes. The Whelan And Goldman (WAG) matrix method was used to estimate a phylogenetic tree with the protein sequences shared by all the selected genomes. Percentual bootstrap values were calculated and represented on the internal branches of the tree. The genomes of *K. oxytoca* KCTC 1686 and *K. oxytoca* E718 were used as outgroups. MLST and KL-typing analyses, in addition to virulence and drug resistance screening, were performed for all 172 genomes as described for strain B31. The dendrograms displaying annotated virulence and resistance information were plotted using iTOL (Interactive Tree Of Life) (Letunic and Bork, 2019). Gegenees (version 2.1) (Agren et al., 2012;7.) was used to set a DNA-based identity matrix among 44 representative genomes selected according to the phylogroups distribution and prevalence of STs. A heatmap plot was generated in agreement with the percentages of DNA similarities. A DNA-based phylogenetic tree was generated using SplitsTree (version 4.14.5) (Kloepper and Huson, 2008) and the identity matrix and neighbor joining methods (Huson and Bryant, 2006).

3. Results

3.1. Clinical aspects of infection and antibiotic resistance

In May 2016, a previously healthy 41-year-old female researcher was admitted to an Intensive Care Unit (ICU) in Brazil presenting dyspnea, systemic fatigue, and fever. Computed tomography scan of the thorax revealed bilateral pneumonia with extensive and diffuse areas of necrosis in the lung parenchyma. Due to the impaired lung function, the patient underwent mechanical ventilation for 14 days. Plating of the bronchoalveolar lavage resulted in the observation of separated bacterial colonies of a single morphological type, which was identified as strain B31. MALDI Biotyper provided the *K. pneumoniae* identification score of 2.35, assigned to highly probable species identification. The patient was considered cured after 10 months of antibiotic cocktail treatment guided by the AST results for strain B31 (Table 1).

Resistance was observed for eight antibiotics, including several beta-lactams (ampicillin, cefepime, ceftazidime, ceftriaxone, cefuroxime axetil, and cefuroxime), one beta-lactamase inhibitor (sulbactam), and one fluoroquinolone (ciprofloxacin). Susceptibility (antibiotic effectiveness) was observed for several beta-lactams (including ceftazidime, ceftriaxone, cefuroxime, and meropenem), aminoglycosides (amikacin and gentamicin), and polymyxin E (colistin). Intermediate resistance, which might occur when the microorganism is located in sites within the host where the drug is not easily accessible (Rodloff et al., 2008), was observed for the combination of ampicillin with tazobactam – a beta-

Table 1

AST results for strain B31.

Antimicrobials	MIC (μg/mL)	Interpretation
Piperacillin ^b /Tazobactam ^c	32	Intermediate
Tigecycline ^f	2	Intermediate
Ampicillin ^b /Sulbactam ^c	≥32	Resistant
Ampicillin ^b	≥32	Resistant
Cefepime ^b	≥64	Resistant
Ceftazidime ^b	≥64	Resistant
Ceftriaxone ^b	≥64	Resistant
Cefuroxime axetil ^b	≥64	Resistant
Cefuroxime ^b	≥64	Resistant
Ciprofloxacin ^d	4	Resistant
Amikacin ^a	4	Susceptible
Cefoxitin ^b	≤4	Susceptible
Colistin ^e	≤0.5	Susceptible
Ertapenem ^b	≤0.5	Susceptible
Gentamicin ^a	≤1	Susceptible
Imipenem ^b	≤0.25	Susceptible
Meropenem ^b	≤0.25	Susceptible

Note: ^aAminoglycoside; ^bBeta-lactam; ^cBeta-lactamase inhibitor; ^dFluoroquinolone; ^ePolymyxin E; ^fTetracycline.

lactamase inhibitor – and for tigecycline, a tetracycline. Based on the antimicrobial resistance findings and the clinical observations, the isolate was considered multidrug resistant (MDR) (Magiorakos et al., 2011) and demonstrated high capacity to establish persistent infections.

In strain B31, the majority of the genes determining resistance to antibiotics are located in two plasmid DNA sequences, one with the replicon IncI1 and the other with IncFII_k and IncFIB_k (Table 2). In addition to three plasmid genes (*bla*_{OXA-1}, *bla*_{CTX-M-15}, and *bla*_{TEM}), two chromosomal genes (*ampH* and *bla*_{SHV}) explain B31 resistance to several beta-lactam antibiotics. In turn, the plasmid genes *mphA* and *catB4* encode the resistance enzymes macrolide phosphotransferase and group B chloramphenicol acetyltransferase, respectively (Shakya and Wright, 2010; Bunney et al., 1995). Resistance of B31 to sulfonamide (Sul), tetracycline (Tet), and trimethoprim (Tmt) is likely conferred by the plasmid genes *sulI*, *tetA*, and *dfrA12*, respectively. Resistance against the fluoroquinolone ciprofloxacin was likely acquired from mutations in the chromosomal genes *gyrA* (DNA gyrase subunit A) and *parC* (DNA topoisomerase 4 subunit A) (Bagel et al., 1999).

Deactivation of the gene *mgrB* has been shown to promote colistin resistance in *K. pneumoniae* (Cannatelli et al., 2014; Poirel et al., 2015). In B31, the *mgrB* allele presents only 49 % of nucleotide coverage over the functional allele deposited in the ARG-ANNOT database (Gupta et al., 2014), meaning it suffered truncation/function loss. Kleborate allowed the identification of two imprecise allele matches related to aminoglycoside resistance (*aadA2*; *aph3-Ia*). Indeed, these alleles did not confer resistance to the aminoglycoside drugs amikacin and gentamicin,

Table 2

Drug resistance genes predicted in the genome of strain B31.

Antibiotic resistance classes	Plasmid genes	Chromosomal genes
AGly	<i>aadA2</i> *; <i>aph3-Ia</i> *	–
Col	<i>mgrB</i> -49%; <i>pmrB</i> -12%	–
Flq	–	<i>parC</i> -80I; <i>gyrA</i> -83F; <i>gyrA</i> -87A
MLS	<i>mphA</i>	–
Phe	<i>catB4</i>	–
Sul	<i>sulI</i>	–
Tet	<i>tetA</i>	–
Tmt	<i>dfrA12</i>	–
Bla	<i>bla</i> _{OXA-1}	<i>bla</i> _{AMPH} *; <i>bla</i> _{SHV-28} *
Bla_ESBL	<i>bla</i> _{CTX-M-15}	–
Bla_broad_inhR	<i>bla</i> _{TEM-30} *	–

Note: * Imprecise allele matches on Kleborate. Resistance classes and genes are reported using the Kleborate nomenclature.

as revealed by the AST. Strain B31 does not harbor resistance genes for fosfomycin, glycopeptides, rifampin, carbapenemases, extended spectrum beta-lactamases with resistance to beta-lactamase inhibitors, and broad-spectrum beta-lactamases.

3.2. Typing and molecular characterization of strain B31

Genome sequencing of strain B31 revealed a single 5.27 Mb circular chromosome (accession CP035929) with 98.79% of ANI over the genome of the type strain DSM 30,104 of *K. pneumoniae* (Table 3). This result reinforced the taxonomic classification of strain B31 as *K. pneumoniae*, since the ANI percentage was greater than the 96 threshold (Goris et al., 2007; Federhen et al., 2016). In addition, the ANI values of B31 against *K. quasipneumoniae* 01A030, *K. quasipneumoniae* subsp. *similipneumoniae* 07A044, and *K. variicola* DSM 15,968 – some of the *K. pneumoniae* closest phylogenetic relatives – were 93.2%, 93.5%, and 94.13%, respectively.

The MLST analyses assigned strain B31 to the Sequence Type 15 (ST15), as 100 % of nucleotide identity over allele 1 was observed for the genes *gapA*, *infB*, *mdh*, *pgi*, *phoE*, *rpoB*, and *tonB*. Also, strain B31 was assigned to the capsular type KL19, since 18 out of the 20 capsular loci presented very high nucleotide identity (greater than 94 %) and coverage (100 %) over the corresponding alleles of this k-type (Supplementary Table 1). The lowest k-type identity percentages reported were 94.69 and 94.86 (for KL19_14 and *wcuF*, respectively), very close to the 95 % threshold for the best confidence matching.

Pathogenic bacteria like *K. pneumoniae* may evolve through the acquisition of resistance islands (RI) and pathogenicity islands (PI), which might be involved in structural features, motility, and pathogen-host interactions (de Soares et al., 2016). Among 33 genomic islands predicted in the genome of B31, 11 were classified as resistance islands (RI), 14 as pathogenicity islands (PI), and 8 miscellaneous islands (MIS), which harbor genes related to both pathogenicity and resistance. Genes encoding the Type VI Secretion System (T6SS) and a CRISPR-Cas system were found in pathogenic islands of B31. In addition, one prophage region of 37.7 Kb in length was detected in the genome of B31. Although 29 proteins are likely encoded by this prophage, the PHASTER score of only 60 suggests that the prophage is incomplete in this genome (Arndt et al., 2016) (Supplementary Table 2).

3.3. Comparative genomics with strain B31

Complete genomes assigned to 172 *K. pneumoniae* strains were retrieved from GenBank for comparative genomics against strain B31. Due to reports of imprecise species identification (Long et al., 2017; Nicolás et al., 2018), verification of all these strains was performed using the ANI method. Divergent identifications were found for six strains, including ATCC 43816 (Supplementary Table 3). For instance, the ANI

results suggested that ATCC 43816 relates to *K. quasipneumoniae* sub-species *similipneumoniae*, as its genome presented 98.8 % of ANI over the genome of 07A044, the reference strain for this species. The phylogenetic estimation using the proteins shared by all strains demonstrated that, along with HKUOPLC, ATCC 43816 forms one clade that is phylogenetically closer to the group of the ANI-validated *K. pneumoniae*, in contrast to the clade formed by the other ANI-divergent strains assessed (YH43, KP5-1, 342, and KPN148) (Fig. 1). *K. variicola* and *K. quasipneumoniae* formed one individual clade each, in agreement with previous studies on *Klebsiella* phylogroups (Holt et al., 2015).

The phylogenetic estimation based on proteins shared by all strains revealed that the ANI-validated *K. pneumoniae* isolates cluster together in accordance with the sequence type distribution (Fig. 2). In addition, the phylogenetic distribution resulting from the alignment of DNA fragments shared by 44 representative *K. pneumoniae* strains, including B31, resembled the sequence type clustering pattern seen on the protein-based phylogeny with all 173 strains (Fig. 4). These results corroborate the ST information reflecting the phylogenetic relationships among several *K. pneumoniae* strains. Separation of strains by country of isolation was not apparent in any of the phylogeny clustering methods applied.

Out of the 173 strains, only three (P1428, RJF999, and CIP_52.145) were found to harbor all virulence genes screened (*ybt*, *clb*, *iuc*, *iro*, and *rmpA/rmpA2*) (Fig. 2). Apart from these strains, only eight (HS11286, KCTC 2242, KP8, NTUH-K2044, RJA166, RJF293, VK055, and KP11) harbor genes for the hypermucoid phenotype (Fig. 2). The serological K-types ranged from KL1 to KL74 for 125 strains, including B31 (Ørskov and Fife-Asbury, 1977). The remaining 48 strains were found to belong to K-type KL101 or greater, which are exclusively defined by gene sequences. Although none strain presented genes coding for extended spectrum beta-lactamases, at least one narrow spectrum beta-lactamase gene (*bla*) was predicted in every strain analyzed. However, narrow spectrum beta-lactamase was the only class of antibiotics resistance predicted in 32 strains (Fig. 3).

Like six other strains, B31 was assigned to ST15 and presented high proximity with groups ST14 and ST383 in the protein-based phylogeny tree (Figs. 2 and 3). Phylogeny analysis with shared DNA sequences revealed one single cluster for STs 14 and 15, with DNA similarities of 90% or greater between strains (Fig. 4). Indeed, STs 14 and 15 formed a single clonal complex (CC15), and the differences between them are restricted to changes in the nucleotide sequence of gene *infB*. In the ST15 group, only four strains (MLST-15, KP36, BR, and PMK1) harbor the

Table 3
K. pneumoniae species confirmation for strain B31 using ANI.

	R1	R2	R3	R4	B31
R1	–	96.12 (87.12)	92.9 (87.15)	93.00 (83.58)	93.2 (84.18)
R2	96.34 (92.46)	–	93.37 (92.11)	93.38 (88.4)	93.5 (88.84)
R3	92.9 (86.63)	93.24 (86.39)	–	94.09 (83.47)	94.13 (84.89)
R4	93.07 (83.04)	93.30 (82.72)	94.15 (83.22)	–	98.79 (85.44)
B31	93.21 (87.74)	93.49 (87.14)	94.13 (89.1)	98.67 (90.04)	–

Note: ANI values above 96% (in bold) indicate strains belonging to same species; values in parentheses represent nucleotide alignments (%). *K. quasipneumoniae* 01A030 (R1), *K. quasipneumoniae* subsp. *similipneumoniae* 07A044 (R2), *K. variicola* DSM 15968 (R3), and *K. pneumoniae* DSM 30104 (R4) were used as the references for species confirmation of B31.

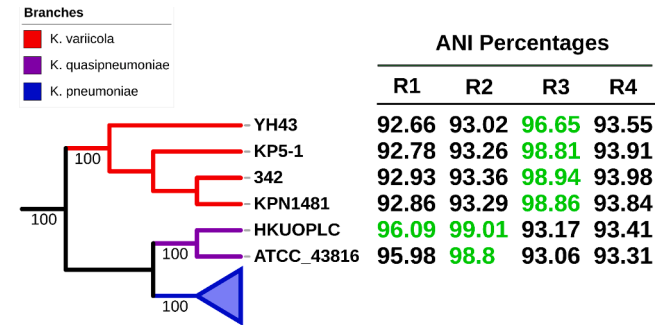


Fig. 1. Representation of the phylogroups formed by the different species found among all *Klebsiella* spp. genomes retrieved for analysis. The *K. pneumoniae* ANI-divergent strains are YH43, KP5-1, 342, KPN1481, HKUOPLC, ATCC 43816. The *K. variicola* (red branches) and *K. quasipneumoniae* (purple branches) strains constituted separated clades. ANI percentages between isolates are shown on the right side of the dendrogram. ANI values greater than 96% are shown in green. Bacteria used as the reference strains are *K. quasipneumoniae* 01A030 (R1), *K. quasipneumoniae* subsp. *similipneumoniae* 07A044 (R2), *K. variicola* DSM 15,968 (R3), and *K. pneumoniae* DSM 30,104 (R4). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

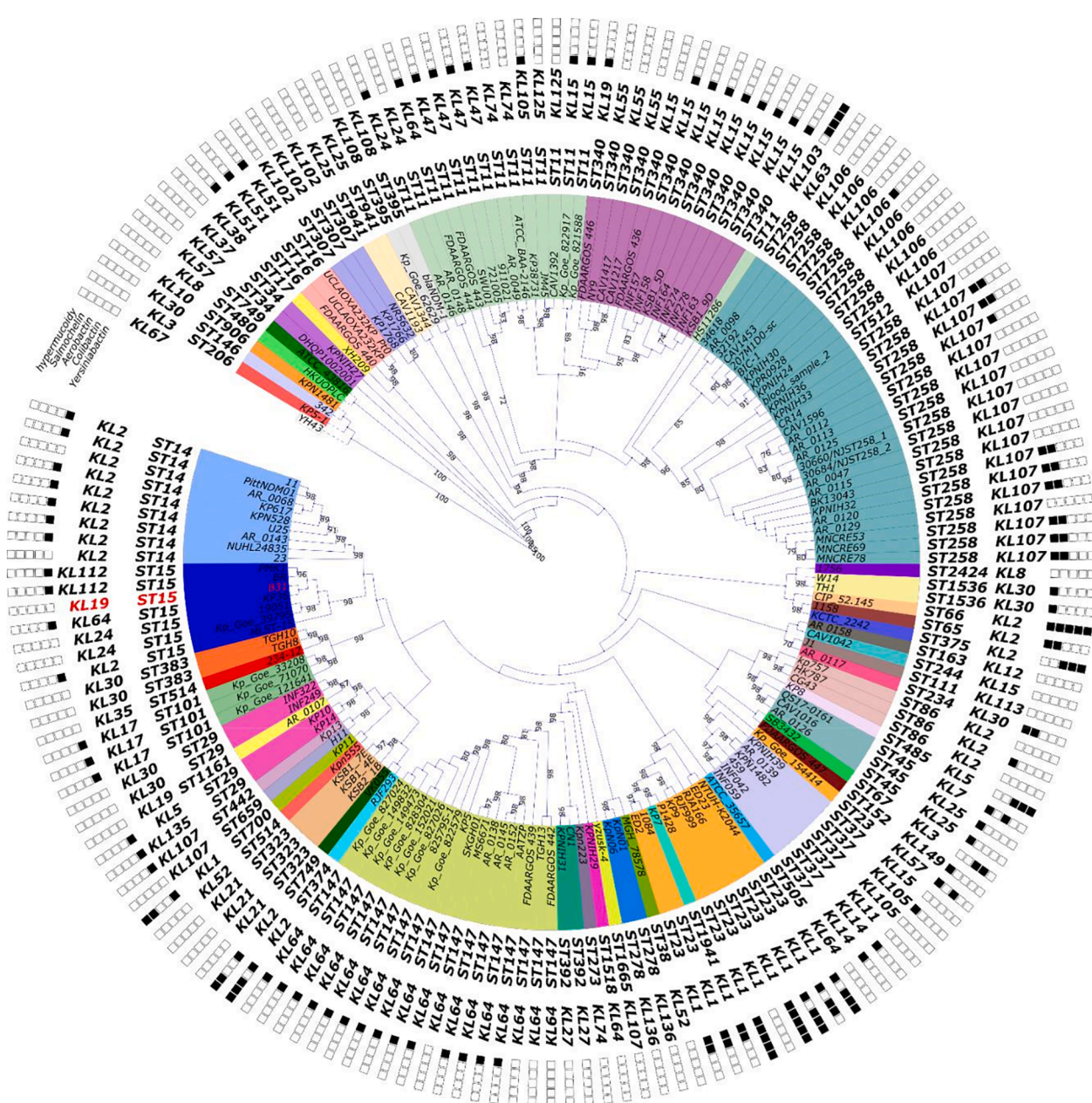


Fig. 2. Protein-based phylogenetic tree comprising 173 *K. pneumoniae* strains annotated with sequence type, capsular type and virulence information. Identification colors allow for distinction of sequence types (STs). Checkboxes filled in black indicate the presence of the virulence genes for yersiniabactin, colistin, aerobactin, salmochelin, and hypermucoidy. Strain B31 and its respective ST and k-type information are highlighted with font color red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

gene coding yersiniabactin siderophore (*ybt*), a high-affinity iron chelator (Fig. 2). All ST15 strains considered in the present study were isolated from humans and most from different parts of the world, including Germany (39795), Nepal (PMK1), China (BR), and Taiwan (KP36). Apart from B31, the only Brazilian isolate used for comparative genomics was Kp13 (Ramos et al., 2014), which is the only strain assigned to ST442. In contrast to B31, strain Kp13 presented closer phylogenetic relationship with STs 29, 1161, 659, and 700. Nevertheless, the MLST analyses revealed that B31 and Kp13 share one allele at the *pgi* locus.

4. Discussion

Emergence of multidrug-resistant bacteria has become a critical public health concern, as the variety of effective antimicrobial agents is

reduced (Anne-Sophie and Benoit, 2017). *K. pneumoniae* is one of the most important pathogens related to nosocomial infections, with antimicrobial resistance playing a crucial role for persistence in healthcare facilities (Alcántar-Curiel et al., 2018). Although the strain reported in the present study has been isolated from a patient admitted to an ICU facility in Brazil, the infection likely occurred in the community, since the symptoms had manifested prior to the hospitalization. In other countries, it has been observed a recent increase in cases of community-acquired infections by *K. pneumoniae* with extended spectrum beta-lactamase resistance, like B31 (Lin et al., 2016). As many bacteria have the potential to cause dyspnea, systemic fatigue, fever, and other airway symptoms, it was important to correctly identify and characterize the causing agent not only to treat the patient, but also to assist policies and procedures to prevent the spread of the microorganism (Lee et al., 2019). Importantly, anamnesis of the patient had raised the

possibility of a laboratory-acquired infection by *Corynebacterium pseudotuberculosis*. Researchers who manipulate pathogenic bacteria in laboratory facilities present increased risk of contamination by these agents (WHO. Laboratory biosafety manual Third edition World Health Organization, 2004).

Although genomics analysis can reveal virulence factors of pathogens (Xu et al., 2018), genomic studies with *K. pneumoniae*, including sequencing, screening, and characterization of virulence and drug resistance determinants, are very scarce in Brazil. One report involved the genomic characterization of Kp13, a carbapenemase resistant *Klebsiella pneumoniae* strain isolated during a nosocomial outbreak in South

Brazil (Ramos et al., 2014). More recently, another study sequenced the genome of ten colistin-resistant strains isolated in two hospitals in Rio de Janeiro state (Longo et al., 2019). In that study, two ST15 strains with resistance to several antibiotics were submitted to whole-genome sequencing and genomic characterization (Longo et al., 2019). In the present study, however, it was not possible to retrieve the genome sequences of these isolates to perform comparative analyses with strain B31. Among the strains included to perform a thorough comparison, there were strain Kp13, assigned to ST442, and more 171 complete genomes retrieved from the GenBank database.

Among the genomes assigned to *K. pneumoniae* and recovered from

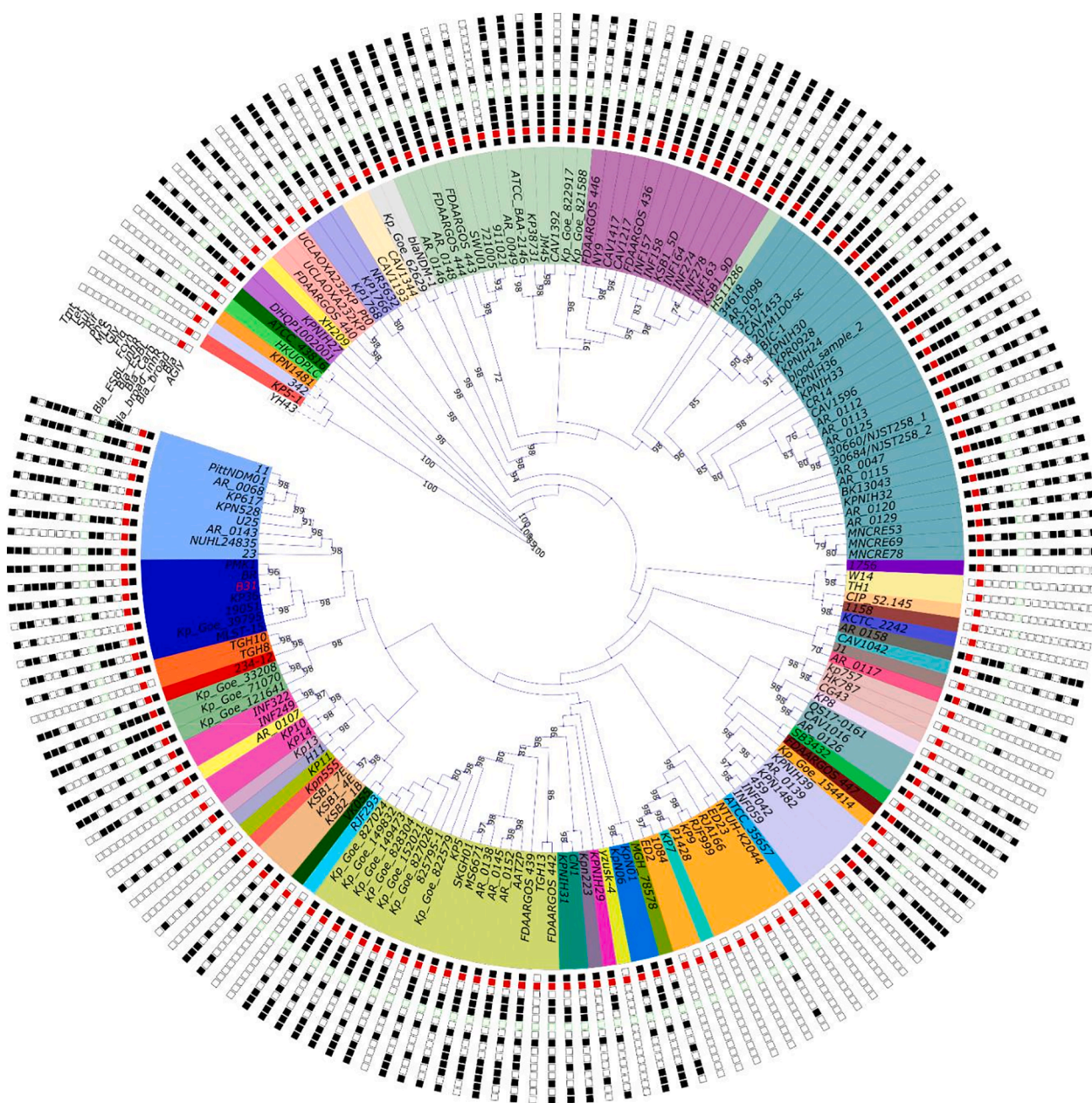


Fig. 3. Protein-based phylogenetic tree annotated with the antibiotic resistance classes. Resistance classes are AGly (aminoglycosides), Bla (beta-lactamases), Bla_broad (broad spectrum beta-lactamases), Bla_broad.inhR (broad spectrum beta-lactamases with resistance to beta-lactamase inhibitors), Bla_Carb (carbapenemases), Bla_ESBL (extended spectrum beta-lactamases), Bla_ESBL.inhR (extended spectrum beta-lactamases with resistance to beta-lactamase inhibitors), Fcyn (fosfomycin), Flq (fluoroquinolones), Gly (glycopeptides), MLS (macrolides), Rif (rifampin), Sul (sulfonamides), Tet (tetracyclines), and Tmt (trimethoprim). Identification colors allow for distinction of sequence types (STs). Checkboxes filled in black indicate the presence of one or more genes related to a specific antibiotic resistance class. Narrow spectrum beta-lactamase gene (*bla*) is present in all strains (checkboxes in red). None strain presented genes for resistance to beta-lactamase inhibitors related to extended spectrum beta-lactamases (checkboxes in green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

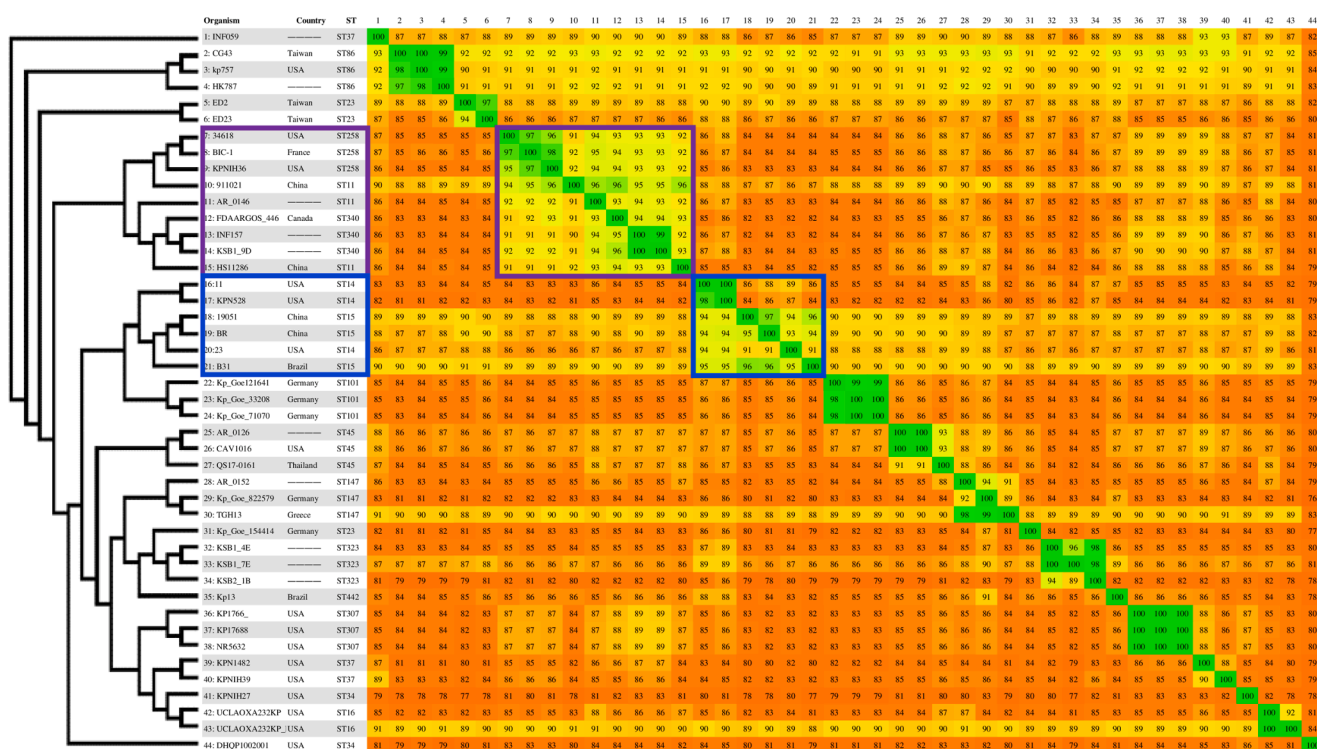


Fig. 4. DNA fragment-based phylogenetic tree with similarity heatmap for 44 strains of *K. pneumoniae*. Heatmap plot ranges from red (lowest similarity) to green (highest similarity). The strain identification numbers shown on the left side (vertically) correlate to the numbers arranged on the top of the plot (horizontally). Two clonal complexes, CC258 and CC15, are highlighted by the purple and blue boxes, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the database, two were found to belong to *K. quasipneumoniae* and four to *K. variicola*, according to the ANI analyses performed. Although not initially expected, this result is consistent with the fact that at least the majority of these genome sequences had been submitted to the NCBI prior to the publication of studies suggesting the use of genomic information to distinguish *K. pneumoniae* from close related species (Long et al., 2017; Gorrie et al., 2017; Martínez-romero and Rodríguez-medina, 2018). Assuming that *K. pneumoniae*; *K. quasipneumoniae* and *K. variicola* form a taxonomic group with no formal designation, whose members show only 90 % of ANI over isolates of other species in the genus, the comparative analyses performed in the present study were performed without discarding them (Wyres et al., 2020).

The phylogenetic analyses revealed two main bacterial clusters formed by two clonal complexes: CC258 (ST258, ST340, and ST11) and CC15 (ST14 and ST15, including B31). CC258 and CC15 were associated with *K. pneumoniae* strains resistant to third-generation cephalosporins in Maghreb (Morocco), West Africa (Senegal, Côte d'Ivoire), Central Africa (Cameroon), East Africa (Madagascar), and Southeast Asia (Vietnam) (Breurec et al., 2013). ST15 isolates have spread globally and are responsible for the high prevalence in Europe and Asia of extended spectrum beta-lactamase resistance, due to the *bla*_{CTX-M-15} gene (Lee et al., 2011; Wyres and Holt, 2016). ST15 strains often present resistance to beta-lactams and fluoroquinolones and are also associated with MCR-1 colistin resistance (Caspar et al., 2017). In numerous reports, ST15 strains presented resistance to carbapenems in the hospital environment (Chung The et al., 2015). ST15 strains have also been associated with infection in companion animals, suggesting their importance to community infections (Marques et al., 2019). Recent isolation of ST15 strains in Brazil reinforces the worldwide spread potential of multidrug-resistant *K. pneumoniae*. Interestingly, strain B31 presented mutations on the plasmid gene *mgrB*, and mutations had also been reported on this gene for two ST15 strains isolated in Rio de Janeiro (Longo et al., 2019). However, unlike B31, these two strains were found to be resistant to

colistin possibly due to mechanisms associated with other resistance genes.

To our knowledge, the present study is the first to sequence and characterize the genome of a *K. pneumoniae* isolated in Northeast Brazil. Noteworthy, several *K. pneumoniae* strains have been isolated in that region and molecularly characterized. For instance, in the state of Pernambuco, 15 ciprofloxacin-resistant *K. pneumoniae* isolates assigned to ST147 were recovered from the urine of hospitalized patients, while in the state of Tocantins, four out of ten MDR isolates were assigned to ST29 (Rocha et al., 2014; Ferreira et al., 2019). In the present study, both ST147 and ST29 formed phylogroups that are closer than the majority of STs to the group of ST15 (Fig. 2), denoting close evolutionary relationship among them in this region of Brazil. In addition, our study stands out for the characterization of one *K. pneumoniae* strain that had likely infected the patient outside healthcare facilities, although the origin of infection remained unclear.

The majority of the drug resistance determinants found in B31 are plasmid driven. *K. pneumoniae* plasmids often carry genes for conjugation, so they can be transferred and incorporated by other bacteria if any other plasmid of the same incompatibility group is present. As one of the plasmids of B31 presented two replicons (IncFII_k and IncFIB_k), chances of spread to other *Enterobacteriaceae* are increased, since the effect of one incompatible replicon in the cell might be neutralized by the presence of a second replicon from another incompatibility group. In fact, IncFII_k plasmids have a role as a reservoir for antibiotic resistance genes in clinical isolates and are largely associated with the dissemination of extended spectrum beta-lactamase resistance (Bi et al., 2018). Noteworthy, B31 also hosts an IncI1 plasmid, and there are reports on plasmids of this incompatibility group carrying several genes for resistance to beta lactams (García-Fernández et al., 2008).

The genes encoding some of the main *K. pneumoniae* virulence factors (yersiniabactin, colibactin, aerobactin and salmochelin) were not found in the genome of B31. The genes *rmpA* and *rmpA2*, which upregulate the

capsule production and cause the hypermucoid phenotype, were not found either. These results suggest that B31 is not a hypervirulent strain. However, other virulence determinants might have contributed to the long persistence of this strain in the patient's organism. Several studies reported that the acquisition of virulence factors through horizontal gene transfer contributes to the invasive diseases caused by *K. pneumoniae* (Ramos et al., 2014; Holt et al., 2015; Caspar et al., 2017; Wyres and Holt, 2018). For example, the type 1 fimbriae genes were found in one pathogenic island (PI1) of B31. In *K. pneumoniae*, fimbriae have been associated with the adherence to the eukaryotic cells and occurrence of urinary tract infections (Struve et al., 2008). As an important virulence factor of pathogens, type VI secretion system (T6SS) plays an important role in infection (Luo et al., 2020). The T6SS genes were also found in the genome of B31. It has been demonstrated that the *K. pneumoniae* T6SS secretes effector proteins, such as toxins, to surpass competing bacteria in the environment (Gallique et al., 2017; Liu et al., 2017). In addition, only a single prophage region was predicted in the genome of B31. This might be explained by the CRISPR-Cas system genes present in one of the genomic islands predicted (PI5). CRISPR-Cas is part of the prokaryotic adaptive immune system in which invading DNA, such as bacteriophage DNA, is recognized and degraded to avoid recombination with the genome (Wiedenheft et al., 2012).

As explored in the present study, comparative genomics may reveal traits shared by several phylogroups or exclusive to a specific phylogroup. The whole-genome sequencing of MDR bacteria such as *K. pneumoniae* has greatly improved the taxonomic classification and understanding of relevant traits such as antimicrobial resistance and virulence. The genetic diversity of *K. pneumoniae* is still barely known in Brazil, and the present study represents a step forward with the first *K. pneumoniae* genome sequence and characterization in Northeast Brazil. Further genome characterization of *K. pneumoniae* in Brazil will contribute to better understand how the strains in the country correlate to each other and to strains in different parts of the world, supporting the development of valuable strategies to mitigate dissemination and infection by this bacterium.

5. Ethics

Written informed consent was obtained from the patient, with protocols approved by and in accordance with the Research Ethics Committee (COEP 5149/2016) in Federal University of Minas Gerais, Belo Horizonte, Brazil.

6. Data, code and materials

DNA sequence: Genbank accession: CP035929

Other datasets supporting this article have been uploaded as part of the [Supplementary Material](#).

7. Authors' contributions

Wrote the manuscript (R.P., N.S., S.T., A.C.C., R.S.); designed the study (R.P., S.T., P.I.P.R., V.A., and T.L.P.C.); experimental work (B.B., N.S., D.H.B., L.T.O. and R.S.); conducted in silico analyses and interpreted the results (R.P., M.V.C.V., A.K.J., R.B.K. and P.I.P.R.); critically reviewed and revised the manuscript (N.S., A.G.G., A.G.N., F.F.P., I.B.L., H.C.P.F., S.C.S., D.B., P.G., R.M., P.I.P.R., and T.L.P.C.); supervised the study (S.T., V.A., and T.L.P.C.). Resources (H.C.P.F., P.G. and B.B.). All authors approved this manuscript for publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gene.2020.145386>.

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